Project: Worked examples
Title: Duct sizing
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Sizes each duct section by the equal-friction method (a target friction rate) or the velocity method (a target velocity), rounds to standard sizes, and reports velocity, friction rate, friction and fitting losses per section and the total pressure drop along the index run, with warnings against user velocity and pressure-drop limits.

Round: D from the target (friction: Colebrook with the air properties at T; velocity: D = √(4Q/πv)), rounded up to the next standard size. Rectangular: equivalent diameter D_e = 1.30 (ab)^0.625/(a + b)^0.25 (equal friction and capacity); flat oval: D_e = 1.55 A^0.625/P^0.25 with A = πb²/4 + b(a − b), P = πb + 2(a − b). Friction rate Δp/L = f ρ v_e²/(2D_e) with v_e the velocity in the equivalent round duct; fitting loss ΣC ρ v²/2 at the actual velocity; totals summed over the index-run sections.

Ref: ASHRAE Handbook — Fundamentals Ch. 21 — Duct design (friction chart basis, equivalent diameters, loss coefficients)
Ref: CIBSE Guide C Section 4 — Flow of fluids in pipes and ducts
Ref: BS EN 1505 — Ventilation for buildings: sheet metal air ducts and fittings with rectangular cross-section, dimensions
Ref: BS EN 1506 — Ventilation for buildings: sheet metal air ducts and fittings with circular cross-section, dimensions

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
method	Sizing method	equal friction		assumption: Equal friction for a low-pressure system.	any
dp_target	Target friction rate (equal friction)	1	Pa/m	spec: SPEC-M-07 §5: 1 Pa/m design friction rate	0.3 … 5 Pa/m
v_target	Target velocity (velocity method)	6	m/s	default	2 … 20 m/s
T	Air temperature	20	degC	assumption: Supply air 20 °C.	-20 … 60 degC
p_atm	Barometric pressure	101.325	kPa	assumption: Sea level.	80 … 105 kPa
eps	Duct roughness	0.09	mm	spec: Galvanised sheet, medium roughness (ASHRAE Ch. 21 category)	0.03 … 3 mm
sections	Duct sections	5 row(s)		drawing: DRG-M-205 rev B	any
v_limit	Velocity limit (warning and check)	8	m/s	spec: SPEC-M-07 §5: 8 m/s in risers	2 … 30 m/s
dp_limit	Total pressure-drop limit (check)	250	Pa	spec: Fan external static allowance 250 Pa	50 … 2000 Pa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
round_to	Round sizes up to a multiple of	50	mm	spec: BS EN 1505 / 1506 preferred sizes, 50 mm steps	1 … 100 mm

3. Assumptions

A1. Air is dry at the stated temperature and pressure (ρ = p/(287.055 T)); viscosity by Sutherland's law. (Standard air properties.)
A2. Rectangular and flat-oval ducts are sized through the equivalent diameter for equal friction and capacity. (Huebscher / Heyt–Diaz relations as given in ASHRAE Fundamentals Ch. 21.)
A3. Fitting loss coefficients C are user-supplied (manufacturer or ASHRAE/CIBSE data, not embedded) and refer to the section's velocity pressure. (Non-negotiable: no standard tables embedded.)
A4. method (Sizing method) is taken as equal friction. (Equal friction for a low-pressure system.)
A5. T (Air temperature) is taken as 20 degC. (Supply air 20 °C.)
A6. p_atm (Barometric pressure) is taken as 101.325 kPa. (Sea level.)

4. Working

[10] rho = p / (287.055 T)
= 101320 Pa / (287.055 × 293.15 K)
= 1.2041 kg/m³
Air density

[11] mu = Sutherland's law
= T = 293.15 K
= 18.133 uPa·s
Air dynamic viscosity

[12] a_s1 = AHU to riser: side a
= as given
= 800 mm
Side a, AHU to riser

[13] b_s1 = AHU to riser: side b
= from the target equivalent diameter 573.91 mm
= 400 mm
Side b, AHU to riser

[14] De_s1 = AHU to riser: equivalent diameter — ASHRAE Ch. 21
= 1.30 (ab)^0.625/(a + b)^0.25
= 609.35 mm
Equivalent diameter, AHU to riser

[15] v_s1 = AHU to riser: Q / A
= 2 / 0.32
= 6.25 m/s
Velocity, AHU to riser

[16] Re_s1 = AHU to riser: □ v_e D_e / □ (v_e = 4Q/πD_e²)
= 1.2041 × 6.8581 × 0.60935 / 0.000018133
= 277500
Reynolds number, AHU to riser

[17] f_s1 = AHU to riser: friction factor
= Colebrook–White
= 0.015999
Friction factor, AHU to riser

[18] dpL_s1 = AHU to riser: f □ v_e² / (2 D_e)
= 0.015999 × 1.2041 × 6.8581² / (2 × 0.60935)
= 0.74347 Pa/m
Friction rate, AHU to riser

[19] dp_f_s1 = AHU to riser: (p/L) L
= 0.74347 × 15
= 11.152 Pa
Friction loss, AHU to riser
[20] dp_fit_s1 = AHU to riser: C v²/2
= 0.8 × 23.518
= 18.814 Pa
Fitting loss, AHU to riser
[21] dp_s1 = AHU to riser: friction + fittings + equipment
= 11.152 + 18.814 + 0
= 29.966 Pa
Section loss, AHU to riser (index run)
[22] D_s2 = riser: diameter
= target 573.91 mm rounded up to 50 mm
= 600 mm
Diameter, riser
[23] v_s2 = riser: Q / A
= 2 / 0.28274
= 7.0736 m/s
Velocity, riser
[24] Re_s2 = riser: v_e D_e / (v_e = 4Q/πD_e²)
= 1.2041 × 7.0736 × 0.6 / 0.000018133
= 281820
Reynolds number, riser
[25] f_s2 = riser: friction factor
= Colebrook–White
= 0.015985
Friction factor, riser
[26] dpL_s2 = riser: f v_e² / (2 D_e)
= 0.015985 × 1.2041 × 7.0736² / (2 × 0.6)
= 0.80253 Pa/m
Friction rate, riser
[27] dp_f_s2 = riser: (p/L) L
= 0.80253 × 12
= 9.6303 Pa
Friction loss, riser
[28] dp_fit_s2 = riser: C v²/2
= 0.3 × 30.124
= 9.0371 Pa
Fitting loss, riser
[29] dp_s2 = riser: friction + fittings + equipment
= 9.6303 + 9.0371 + 0
= 18.667 Pa
Section loss, riser (index run)
[30] a_s3 = floor main: side a
= as given
= 500 mm
Side a, floor main
[31] b_s3 = floor main: side b
= from the target equivalent diameter 473.16 mm
= 400 mm
Side b, floor main

[32] De_s3 = floor main: equivalent diameter ASHRAE Ch. 21
= 1.30 (ab)^0.625/(a + b)^0.25
= 488.12 mm
Equivalent diameter, floor main
[33] v_s3 = floor main: Q / A
= 1.2 / 0.2
= 6 m/s
Velocity, floor main
[34] Re_s3 = floor main: ρ v_e D_e / μ (v_e = 4Q/πD_e²)
= 1.2041 × 6.4127 × 0.48812 / 0.000018133
= 207850
Reynolds number, floor main
[35] f_s3 = floor main: friction factor
= Colebrook–White
= 0.016905
Friction factor, floor main
[36] dpL_s3 = floor main: f ρ v_e² / (2 D_e)
= 0.016905 × 1.2041 × 6.4127² / (2 × 0.48812)
= 0.85744 Pa/m
Friction rate, floor main
[37] dp_f_s3 = floor main: (Δp/L) L
= 0.85744 × 25
= 21.436 Pa
Friction loss, floor main
[38] dp_fit_s3 = floor main: ΣC ρ v²/2
= 1.1 × 21.674
= 23.841 Pa
Fitting loss, floor main
[39] dp_s3 = floor main: friction + fittings + equipment
= 21.436 + 23.841 + 0
= 45.277 Pa
Section loss, floor main (index run)
[40] D_s4 = branch to VAV: diameter
= target 312.82 mm rounded up to 50 mm
= 350 mm
Diameter, branch to VAV
[41] v_s4 = branch to VAV: Q / A
= 0.4 / 0.096211
= 4.1575 m/s
Velocity, branch to VAV
[42] Re_s4 = branch to VAV: ρ v_e D_e / μ (v_e = 4Q/πD_e²)
= 1.2041 × 4.1575 × 0.35 / 0.000018133
= 96625
Reynolds number, branch to VAV
[43] f_s4 = branch to VAV: friction factor
= Colebrook–White
= 0.01938
Friction factor, branch to VAV
[44] dpL_s4 = branch to VAV: f ρ v_e² / (2 D_e)
= 0.01938 × 1.2041 × 4.1575² / (2 × 0.35)
= 0.57623 Pa/m
Friction rate, branch to VAV

[45] dp_f_s4 = branch to VAV: (Δp/L) L
= 0.57623 × 10
= 5.7623 Pa
Friction loss, branch to VAV
[46] dp_fit_s4 = branch to VAV: ΣC × v²/2
= 0.9 × 10.406
= 9.3657 Pa
Fitting loss, branch to VAV
[47] dp_s4 = branch to VAV: friction + fittings + equipment
= 5.7623 + 9.3657 + 60
= 75.128 Pa
Section loss, branch to VAV (index run)
[48] D_s5 = other branch: diameter
= target 406.05 mm rounded up to 50 mm
= 450 mm
Diameter, other branch
[49] v_s5 = other branch: Q / A
= 0.8 / 0.15904
= 5.0301 m/s
Velocity, other branch
[50] Re_s5 = other branch: ρ v_e D_e / μ (v_e = 4Q/πD_e²)
= 1.2041 × 5.0301 × 0.45 / 0.000018133
= 150310
Reynolds number, other branch
[51] f_s5 = other branch: friction factor
= Colebrook–White
= 0.017809
Friction factor, other branch
[52] dpL_s5 = other branch: f × ρ v_e² / (2 D_e)
= 0.017809 × 1.2041 × 5.0301² / (2 × 0.45)
= 0.60286 Pa/m
Friction rate, other branch
[53] dp_f_s5 = other branch: (Δp/L) L
= 0.60286 × 18
= 10.851 Pa
Friction loss, other branch
[54] dp_fit_s5 = other branch: ΣC × v²/2
= 1 × 15.233
= 15.233 Pa
Fitting loss, other branch
[55] dp_s5 = other branch: friction + fittings + equipment
= 10.851 + 15.233 + 60
= 86.084 Pa
Section loss, other branch
[56] dp_index = Σ section losses on the index run
= 4 section(s)
= 169.04 Pa
Index-run pressure drop
[57] dp_all = Σ section losses, all sections
= 5 section(s)
= 255.12 Pa
Sum over every section (not a path)

[58] v_peak = highest section velocity
=
= 7.0736 m/s
Highest velocity
[59] u_dp = dp_index / dp_limit
= 169.04 / 250
= 0.67616
Pressure-drop utilisation
[60] u_v = v_peak / v_limit
= 7.0736 / 8
= 0.88419
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
a_s1	800 mm	a, AHU to riser
b_s1	400 mm	b, AHU to riser
De_s1	609.35 mm	D_e, AHU to riser
v_s1	6.25 m/s	v, AHU to riser
Re_s1	277500	Re, AHU to riser
f_s1	0.015999	f, AHU to riser
dpL_s1	0.74347 Pa/m	Δp/L, AHU to riser
dp_f_s1	11.152 Pa	Friction loss, AHU to riser
dp_fit_s1	18.814 Pa	Fitting loss, AHU to riser
dp_s1	29.966 Pa	Section loss, AHU to riser
D_s2	600 mm	D, riser
v_s2	7.0736 m/s	v, riser
Re_s2	281820	Re, riser
f_s2	0.015985	f, riser
dpL_s2	0.80253 Pa/m	Δp/L, riser
dp_f_s2	9.6303 Pa	Friction loss, riser
dp_fit_s2	9.0371 Pa	Fitting loss, riser
dp_s2	18.667 Pa	Section loss, riser
a_s3	500 mm	a, floor main
b_s3	400 mm	b, floor main
De_s3	488.12 mm	D_e, floor main
v_s3	6 m/s	v, floor main
Re_s3	207850	Re, floor main
f_s3	0.016905	f, floor main
dpL_s3	0.85744 Pa/m	Δp/L, floor main

SYMBOL	VALUE	DESCRIPTION
dp_f_s3	21.436 Pa	Friction loss, floor main
dp_fit_s3	23.841 Pa	Fitting loss, floor main
dp_s3	45.277 Pa	Section loss, floor main
D_s4	350 mm	D, branch to VAV
v_s4	4.1575 m/s	v, branch to VAV
Re_s4	96625	Re, branch to VAV
f_s4	0.01938	f, branch to VAV
dpL_s4	0.57623 Pa/m	Δp/L, branch to VAV
dp_f_s4	5.7623 Pa	Friction loss, branch to VAV
dp_fit_s4	9.3657 Pa	Fitting loss, branch to VAV
dp_s4	75.128 Pa	Section loss, branch to VAV
D_s5	450 mm	D, other branch
v_s5	5.0301 m/s	v, other branch
Re_s5	150310	Re, other branch
f_s5	0.017809	f, other branch
dpL_s5	0.60286 Pa/m	Δp/L, other branch
dp_f_s5	10.851 Pa	Friction loss, other branch
dp_fit_s5	15.233 Pa	Fitting loss, other branch
dp_s5	86.084 Pa	Section loss, other branch
dp_index	169.04 Pa	Index-run Δp
dp_all	255.12 Pa	Σ all sections
v_peak	7.0736 m/s	Highest velocity

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Velocity ≤ limit	0.88419	1	0.884	
PASS	Index-run Δp ≤ limit	0.67616	1	0.676	

7. Applicability

No warnings.

8. Verification

Template bs.duct-sizing@1.0.0 Content hash 1d090889cef32044 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0